

New challenges for an NSBI measurement of κ_λ in $HH \rightarrow b\bar{b}\tau\tau$, relative to the first NSBI measurement (ATLAS off-shell $H \rightarrow ZZ \rightarrow 4\ell$)

working note — draft for discussion

20 June 2026

Scope & one-line conclusion

This compares the systematics and background handling of the **first full NSBI measurement** (ATLAS off-shell $H \rightarrow ZZ \rightarrow 4\ell$; note HIGG-2023-01 = arXiv:2412.01600, Sandesara *et al.*) with what a κ_λ **measurement in $HH \rightarrow b\bar{b}\tau\tau$** must do (baselined on CMS AN-18-121, Amendola *et al.*). The NSBI *skeleton* transfers directly; the new difficulty is almost entirely in the *inputs*: $b\bar{b}\tau\tau$ has a far messier final state, several **data-driven backgrounds with no per-event MC density** (QCD, jet $\rightarrow \tau$ fakes), an **irreducible same-final-state single-Higgs background**, an order of magnitude more (and larger) **experimental systematics that deform the very observable (m_{HH}) carrying the κ_λ signal**, and a κ_λ -dependent theory uncertainty. None of these are NSBI-specific — they hit any method — but several sit awkwardly inside a per-event density-ratio framework in ways the clean, all-MC off-shell analysis never faced. Page references to both notes are given inline.

1 The two measurements and the shared NSBI skeleton

ATLAS off-shell [ATLAS p.1,5]: measures the off-shell signal strength $\mu_{\text{off-shell}}$ (hence the Higgs width) in $gg \rightarrow (H^* \rightarrow) ZZ \rightarrow 4\ell$, using 139 fb^{-1} . The physics is a *coupling that morphs an interference shape* (triangle–box in $ggZZ$). **CMS $HH \rightarrow b\bar{b}\tau\tau$** [CMS p.3,4]: searches for non-resonant HH via $ggF+VBF$ in $b\bar{b}\tau\tau$ ($\mathcal{B} = 7.3\%$), 137.6 fb^{-1} , three $\tau\tau$ channels ($\tau_e\tau_h, \tau_\mu\tau_h, \tau_h\tau_h$, plus boosted), extracting κ_λ (and k_{2V}, k_V). $\sigma_{ggF}^{HH} = 31.05^{+2.2\%}_{-5.0\%} \text{ fb}$, $\sigma_{VBF}^{HH} = 1.726 \text{ fb}$ [CMS p.4].

What transfers (the skeleton). The ATLAS recipe (Sec.5) is reusable wholesale: a reference-hypothesis P_{ref} so all NNs estimate ratios to a fixed point; a search-oriented *mixture model* for signal/interference/background; nuisance parameters in both the rate and the per-event density; a profile-likelihood test statistic in a custom JAX+IMINUIT fit; and an interference-aware impact definition [ATLAS p.89,90]. The off-shell interference structure $(\mu - \sqrt{\mu})S + \sqrt{\mu}\text{SBI} + (1 - \sqrt{\mu})B$ [ATLAS p.81] maps onto the κ_λ morphing basis $\{\kappa_\lambda^2, \kappa_\lambda, 1\}$ (since $gg \rightarrow HH = \kappa_\lambda \cdot \text{triangle} + \text{box}$). *The challenges below are about the inputs to this skeleton, not the skeleton itself.*

2 Physics / parameter-of-interest challenges

- **The POI lives in a shape, and that shape is m_{HH} .** κ_λ sensitivity is concentrated in the m_{HH} spectrum (triangle–box interference), strongest in the resolved regime $m_{HH} \in [250, 400] \text{ GeV}$. CMS encodes the κ_λ dependence by *reweighting* the signal MC — a 2D ($m_{HH}, |\cos\theta^*|$) LO reweighting and a dedicated NLO reweighting for the final extraction [CMS p.6]. In NSBI this becomes the $R(x|\kappa_\lambda)$ factor.

- **Multiple correlated POIs / EFT directions.** Beyond κ_λ the analysis also probes k_{2V} , k_V and HEFT operators (c_2, c_{2g}, c_g) [CMS p.4]; the off-shell measurement is effectively one POI. More POIs $\Rightarrow$ a higher-dimensional morphing.
- **k_{2V} sensitivity lives in the high- m_{HH} VBF tail — where binning hurts most.** The $HHVV$ contact term grows with energy unless $k_{2V} = 1$, so VBF-HH k_{2V} sensitivity is concentrated in the high- m_{HH} tail; current analyses use only ~ 4 – 5 bins there. This is precisely the regime where coarse binning discards the most information, so the unbinned NSBI gain is largest on the k_{2V} (and the low- m_{HH} κ_λ) shape directions. (k_{2V} is now the best-measured “inaccessible” coupling — the 2026 ATLAS+CMS combination gives $k_{2V} \in [0.73, 1.3]$.)
- **Interference impact ambiguity (shared, but worse).** Both have interference, so the finite-difference NP-impact definition is ambiguous; ATLAS replaces it with a local implicit-function-theorem impact [ATLAS p.90]. κ_λ needs the same fix.

3 Final state and reconstruction

The off-shell 4ℓ final state is *exquisitely clean*: four leptons, fully reconstructed, with tiny energy-scale systematics and no jets/MET/ τ / b -tagging in the core. $b\bar{b}\tau\tau$ is the opposite:

- **b -jets** (JES/JER, b -tagging) + **hadronic τ** (τ -ID, τ energy scale) + **MET** (the τ neutrinos). The $\tau\tau$ mass is *not* fully reconstructed — it requires an algorithm (SVfit); shifting τ energy shifts MET *and* SVfit coherently [CMS p.93].
- **Three channels** with different background compositions and systematics, plus a boosted regime.

Consequence: the observable x for $b\bar{b}\tau\tau$ is softer and carries many more systematic “handles” than 4ℓ .

4 Background handling — the largest qualitative difference

The crux

Off-shell backgrounds are **all MC-based and clean** (one dominant irreducible $q\bar{q}ZZ$ continuum, modelled in simulation, plus EW). $b\bar{b}\tau\tau$ backgrounds are a **zoo**, **several of them data-driven**, and *data-driven backgrounds have no per-event MC probability density* — which is the object NSBI needs. This is the single hardest conceptual mismatch between $b\bar{b}\tau\tau$ and the off-shell NSBI template.

ATLAS off-shell (Sec. 2.5, Sec. 5). The dominant background is the irreducible $q\bar{q} \rightarrow ZZ \rightarrow 4\ell$ continuum, which *interferes* with the signal and is carried in the mixture model as a component $\mu_{q\bar{q}ZZ}$ (three floating NPs) constrained *in situ* by a dedicated n_{jets} discriminant in a control region [ATLAS p.81]. Plus a small EW component. Everything is MC, so every process has a per-event density the NSBI ratios can target.

CMS $b\bar{b}\tau\tau$ (Sec. 7). A heterogeneous set, with the two largest estimated *from data*:

- **QCD multijet — data-driven ABCD** [CMS p.61,62]. Four regions in (charge $\times$ τ -isolation); shape from region C (data minus other-MC), yield from $C \times B/D$. The QCD *normalisation* uncertainty ranges **5–100%** by category (Table 30, [CMS p.92]); its *shape* uncertainty (alternative B-vs-C templates, symmetrised) is **one of the highest-ranking nuisances** in the most sensitive $\tau_h\tau_h$ channel [CMS p.91,96,97].

- **Drell–Yan (Z +jets) — MC normalised by 18 data-driven scale factors** [CMS p.63]. A simultaneous fit of $M(\mu\mu)$ across **18 control regions** (3 b -jet multiplicities $\times$ 6 Z - p_T bins), giving 18 normalisation SFs (~ 0.6 – 1.7) *and* 18 shape templates (scaled $\pm$) propagated with their covariance [CMS p.65,94].
- **$t\bar{t}$ — MC shape + data-driven normalisation SF** [CMS p.79]. A dedicated CR (inverted elliptical (m_{HH}) mass cut, Eq.22) with $< 7\%$ contamination; COMBINE fit with Barlow–Beeston gives $t\bar{t}$ SF ≈ 0.91 – 0.99 [CMS p.81,82]; the shape is taken from MC (no shape dependence observed).
- **Jet $\rightarrow \tau$ fakes — data-driven** [CMS p.94,95]. $> 93\%$ of the events with a jet faking a τ ; fake-rate uncertainty per region (barrel 24/13/11%, endcap 28/18/25% by year, Table 33), applied as an event weight.
- **Single Higgs — irreducible, same final state** [CMS p.82]. ggH , $VBFH$, VH , $t\bar{t}H$ with $H \rightarrow b\bar{b}$ or $\tau\tau$ mimic the signal; modelled from MC. (Note: $t\bar{t}H/VH$ couplings make this a *coupling-dependent* background.)
- **Di-/tri-boson, W +jets, single- t , V +2jets — MC** [CMS p.82].

Why this is a new, NSBI-specific difficulty

1. **No density for data-driven backgrounds.** QCD (ABCD) and jet $\rightarrow \tau$ fakes are estimated as *yields/weights from data control regions*, not as simulated events with a per-event probability. NSBI’s per-event likelihood ratio $P_X(x)/P_{\text{ref}}(x)$ is undefined for them. Folding an ABCD/fake estimate into an *unbinned* per-event likelihood is an open problem the off-shell analysis never had to solve (its one big background was MC). Options (all imperfect): treat QCD/fakes only in a binned auxiliary term; build a surrogate density from the control-region data; or restrict NSBI to the MC-modelled part and bin the rest.
2. **The QCD shape systematic is a top-ranked nuisance** [CMS p.91] — and it is intrinsically *asymmetric/ill-defined* (one alternative template used for both up and down), forcing a symmetrisation hack [CMS p.96,97]. Carrying that into a per-event morphing is non-trivial.
3. **Single Higgs is an irreducible, coupling-dependent background** in the same final state — it must be modelled *and* morphed consistently with the signal couplings, with its own (correlated) theory systematics.
4. **Many background-normalisation NPs are themselves data-fit results** (18 DY SFs with a covariance, $t\bar{t}$ SF, QCD B/D) — these are constrained *in situ* in the binned CMS fit; reproducing that constraint inside an unbinned NSBI fit requires the CRs to be represented in the per-event likelihood too.

A workaround: flow-based data-driven backgrounds (ABCDnn)

The “no per-event density” problem is the sharpest objection to NSBI here, but it is *not* a dead end — there is a demonstrated solution. **ABCDnn** (Choi, Lim, Oh, arXiv:2008.03636) generalises the ABCD method with a *neural autoregressive flow* that learns the full multi-dimensional CR $\rightarrow$ SR transformation, conditioned on the discriminating observables. Where standard ABCD gives only a scalar SR yield ($N_D = N_B N_C / N_A$), ABCDnn gives a full *differential* background distribution $p_{\text{QCD}}(x)$ — which can then be turned into a density ratio $r_{\text{QCD}}(x) = p_{\text{QCD}}(x)/p_{\text{ref}}(x)$ by training a classifier between flow-generated events and the pooled reference, so it enters the NSBI likelihood

on the same footing as the MC-derived ratios.

The concrete production example (S. An, Dutta/Paulini, CMU PhD thesis 2025, Ch. 5): the CMS *all-hadronic four-top* search, Run-2 (137 fb^{-1} , 13 TeV) + Run-3 (62 fb^{-1} , 13.6 TeV), reaching an **observed (expected) significance** of $3.95 (1.65) \sigma$, $\mu = 2.7^{+1.1}_{-0.9}$. Its single most important design choice is a **yield/shape split** of the dominant $t\bar{t}$ +jets and QCD backgrounds:

- **Yield** $\leftarrow$ **ExtendedABCD**: standard ABCD in (N_{jets}, N_b) *fails* here (the factorisation breaks at high multiplicity — off by $\sim 18\text{--}19.5\%$ in closure); adding two extra sidebands restores it to $\sim 1\text{--}7\%$ (their $\hat{F}_D = (F_B F_C / F_A)^2 (F_X / F_B F_Y)$, Table 5.1–5.2). The lesson: *get the normalisation from a robust closed-form estimator with enough sidebands, not from the flow.*
- **Shape** $\leftarrow$ **ABCDnn** (a neural autoregressive flow): it learns a transformation that *morphs the $t\bar{t}$ MC into the data-driven $t\bar{t}$ +QCD estimate* (target = data – signal – minor bkg per CR), trained by minimising **Maximum Mean Discrepancy** and *conditioned on the CR position* (N_{jets}, N_b) , then applied to $t\bar{t}$ MC in the SR. Notably it never uses QCD MC (too few SR events) — the flow turns pure- $t\bar{t}$ MC into an effective $t\bar{t}$ +QCD mixture.

Three training tricks they found necessary (worth copying): **loss-weighted CR sampling** (sample harder CRs more often, $\propto e^{\text{loss}}$, so every region is learned uniformly); **continuous** (N_{jets}, N_b) **conditioning** instead of one-hot labels, so the flow *interpolates smoothly* across CRs — this is what makes the CR $\rightarrow$ SR *extrapolation* well-behaved; and a **warm-up** (pre-train near identity, RealNVP-style) for stable convergence.

Application to $b\bar{b}\tau\tau$: the **jet $\rightarrow\tau$ fakes** are the natural target — a fake-factor method is conceptually an ABCD with τ -ID inversion as the second axis, so an ABCDnn conditioned on (p_T^τ, η^τ) could learn the full fake- τ kinematic shape instead of the current coarse 2D fake-factor binning; the QCD multijet is the other target; and for $t\bar{t}$ the flow can serve as a data-driven shape cross-check. Following the four-top lesson, keep the **yield/shape split**: take the fake- τ /QCD *normalisation* from the existing fake-factor/ABCD machinery (which CMS already trusts, [CMS p. 61, 94]) and use ABCDnn only for the differential *shape*, then convert that shape to a per-event ratio $r(x)$ for the likelihood. A fully realised $b\bar{b}\tau\tau$ -NSBI would use *ABCDnn shapes (with ABCD/fake-factor yields) for QCD/fakes, MC ratios for the MC-modelled components (Z +jets, $t\bar{t}$, single- H , diboson), and one unbinned NSBI likelihood combining both* — removing the main caveat that NSBI “depends on MC where MC is wrong.”

But the generic caveats remain: (i) *extrapolation distance* — the CR $\rightarrow$ SR map strains for tight SRs far from the sidebands in tagger/isolation space (multiple overlapping CRs validate it but cost stats); (ii) *uncertainty propagation* — the four-top analysis does *not* bootstrap the flow; it splits the uncertainty into three data-driven nuisances (thesis Sec. 6.1, Table 6.1): the *finite-CR-statistics* piece is a $\ln N$ normalisation NP propagated through the ABCD formula (“statistics of extended-ABCD,” 4–26%); the *normalisation-modelling* piece is a $\ln N$ NP from a validation-region (VR) data-vs-prediction closure (weighted-mean deviation $1 - \langle f \rangle + \text{weighted RMS of } f = N_{\text{data}}/N_{\text{pred}}$, 6–44%); and the *ABCDnn shape* uncertainty is a shape NP implemented as a *linear shift of the discriminant*, with the shift size chosen by *minimising a quantitative data/pred-agreement metric* in the VR (floor 1%, correlated across H_T , uncorrelated across years). The whole *uncertainty-estimation* procedure is itself validated in a dedicated orthogonal “closure-test region” that checks pred+syst covers truth in the SR. *Caveat for NSBI*: this VR-closure shape nuisance (“shift the discriminant”) is a *binned-fit* construct; in an unbinned NSBI it becomes yet another per-event morphing nuisance on the ratio $r(x)$ — i.e. the same learned-morphing problem, now applied to the data-driven component; (iii) *signal contamination in CRs* — if signal leaks into the control region the flow learns “background + signal” and biases the SR low, requiring iterative CR definitions with signal subtraction (this caveat is *mild* for $b\bar{b}\tau\tau$: the HH signal is tiny in the fake- τ /QCD

control regions, just as the SM four-top signal was negligible in its CRs); (iv) *no theory handle* — the flow gives a shape but no ME+PS+PDF decomposition, so generator-level theory uncertainties on the background cannot be propagated and are replaced by a data-driven CR→SR closure systematic.

The one caveat specific to using ABCDnn for NSBI (the real gap). In the four-top thesis the flow models only a *low-dimensional* target — the event-level BDT discriminant plus H_T (two variables) — because the downstream fit is a *binned template* fit of that discriminant. NSBI needs the opposite: the *full per-event density* over the whole feature vector, so the data-driven shape can become a per-event ratio $r(x)$. So the four-top result proves flows can deliver a data-driven background *shape of a summary observable*; extending that to the full-dimensional per-event density an unbinned NSBI consumes is a genuine step beyond what is demonstrated — a higher-dimensional flow, harder to train and validate. The conservative intermediate is to let ABCDnn produce the data-driven shape of the *NSBI discriminant* (the per-event log-ratio summed over components), i.e. treat QCD/fakes as a binned component of an otherwise-unbinned likelihood — exactly the “keep QCD binned, NSBI-fy the MC components” compromise.

Net: ABCDnn downgrades the data-driven-background problem from “open” to “solved-with-caveats” for a binned discriminant shape; the fully-unbinned version is a worthwhile but unproven sub-project for a $b\bar{b}\tau\tau$ -NSBI pilot.

5 Systematics handling

What is genuinely shared (equal for both, and equally fallible): the *sources and $\pm 1\sigma$ sizes* of the NPs; the *object-level propagation* (shift the object and all dependents — e.g. $\tau \Rightarrow \text{MET} \Rightarrow \text{SVfit}$ — and recompute the discriminant, [CMS p.93]); the *factorisation over NPs and $\pm 1\sigma$ interpolation*; *Gaussian-constrained profiling*; and the κ_λ -*dependent theory* treatment. ATLAS builds the per-event systematic weight as $P_X(x, \alpha) = w_X(x, \alpha) P_X(x, 0)$ with $w_X(x, \alpha) = \prod_m w_X(x, \alpha_m)$, each single-NP weight interpolated poly-exp between $\pm 1\sigma$ classifier estimates [ATLAS p.83,84]. CMS does the per-*bin* analogue (re-run the DNN on varied inputs → template).

What is harder for $\kappa_\lambda/b\bar{b}\tau\tau$:

1. **κ_λ -systematic shape degeneracy.** The dominant experimental systematics (JES, τ ES, MET, signal parton shower) deform *the same m_{HH} shape* that carries κ_λ . In 4ℓ the big systematics (theory) are largely *separable* from the signal shape; in $b\bar{b}\tau\tau$ they are not. Decorrelation is therefore impossible (it would erase the signal), forcing the “model-it” route — so the accuracy of the morphing $\Delta(x|\alpha)$ *directly sets* the κ_λ uncertainty. (A Fisher-information toy gives, schematically, a $\sim 13\times$ systematic inflation from m_{HH} alone collapsing to $\sim 1.3\times$ with the full feature set — the degeneracy is broken by dimensionality, not avoided.)
2. **An order of magnitude more, and larger, experimental NPs** [CMS p.90–97]: JES split into a reduced set of **11 sources** with a year-correlation scheme [CMS p.95]; **τ ES** p_T -dependent with 4 decay-mode components [CMS p.94]; **b -tag** 5 components (hf, lf, hfstats, lfstats, cferr) [CMS p.96]; **DeepTau** vsJet (5 p_T bins) + vsEle (2 η bins) [CMS p.95,96]; trigger SFs, JER, PU-jet-ID, L1 prefiring, $e/\mu \rightarrow \tau$ fake ES. The off-shell analysis has comparatively few, small (lepton) experimental NPs. Each NP is another per-event weight to learn (~ 2000 NNs already in off-shell = $2 \times \text{NP} \times \text{process}$; $b\bar{b}\tau\tau$ grows this).
3. **Data-driven background systematics that don’t fit the density-ratio picture:** the QCD norm (5–100%) and high-ranking QCD shape [CMS p.91,92,96,97], the 18 DY SF shape/norm templates [CMS p.94], the jet→ τ fake-rate uncertainties (11–28%, Table 33) [CMS p.95], the

VBF dipole-recoil PS uncertainty (20–70%, Table 32) [CMS p.93]. These are template/weight constructs, not deformations of an MC density.

4. **κ_λ -dependent theory uncertainty (POI $\times$ systematic entanglement).** The HH scale/PDF/ m_t uncertainties are folded into a normalisation NP $\text{THU}_{\text{HH}} = +4\% / -18\%$ at the SM point that becomes a κ_λ -dependent uncertainty across the scan [CMS p.91]. The off-shell μ does not entangle with its theory uncertainty this way. Capturing it requires non-factorisable (POI-coupled) morphing.
5. **Factorisation is more likely to break.** $w_X = \prod_m w_X(\alpha_m)$ assumes NPs are independent sources [ATLAS p.83]; but JES and τ ES *both* feed m_{HH} ($= m(b\bar{b} + \tau\tau)$), so a genuine cross-term exists in $b\bar{b}\tau\tau$ that is benign in clean 4ℓ .
6. **MC-statistical uncertainty of the learned ratios.** ATLAS propagates it as one NP via the NN-ensemble spread [ATLAS p.86,87]; CMS uses Barlow–Beeston per bin. With far smaller *systematic-variation* samples in $b\bar{b}\tau\tau$, this term is more important (the nominal MC, $\approx 500\text{k}$ – $750\text{k}/\kappa_\lambda$ -point, is ample for training; the binding constraint is the least-populated $\pm 1\sigma$ sample).

Two systematics that NSBI actually handles *better* than templates. Worth stating the upside too: (a) *parton-shower / generator* uncertainties (Pythia vs. Herwig, ISR/FSR) are a natural fit — a **CARL/DCTR**-style reweighting network learns the per-event weight between alternative showers, which is then profiled as a nuisance; this is strictly more powerful than the current “shape envelope over generator variations” and is the in-likelihood cousin of the RS3L robust-representation idea. (b) *b*-tag / τ -ID scale factors can be *absorbed* into the density ratio and calibrated in situ against a standard candle (the $VZ(c\bar{c})/Z \rightarrow b\bar{b}$ peak trick), turning the POG SF program into an external cross-check rather than the primary constraint. So the systematics ledger is not all cost: *the morphing-heavy ones (JES/ τ ES on m_{HH}) are the burden; the modelling ones (shower, flavour-tagging) are where NSBI’s per-event treatment is an asset.*

6 Statistics and inference

- **Data signal yield is tiny and intrinsic.** $\sigma \times \mathcal{B} \sim 2.5 \text{ fb} \Rightarrow \mathcal{O}(10\text{--}100)$ selected signal events (Run-2 to HL-LHC), under large backgrounds — vs the off-shell tail at $\sim 15\%$ of σ . This is unchanged by MC and keeps the measurement systematics-comparable. (It is also the regime where NSBI’s per-event information helps *most*.)
- **MC/training stats are ample** ($\approx 500\text{k}$ – 750k per κ_λ point), so the morphing/ratio NNs are well-trained — the “under-determined morphing” worry applies to the *systematic-variation* samples, not the nominal.
- **Validation & coverage.** CMS inherits decades-validated COMBINE (asymptotics + toys + Barlow–Beeston). NSBI needs a custom JAX/IMINUIT fit with an explicit Neyman construction and ratio-quality diagnostics (C2ST-like) built and validated from scratch [ATLAS p.64,89].

7 Synthesis: could the NSBI κ_λ result be *worse* than the CMS fit?

Reading what CMS *actually* does removes a strawman: CMS does **not** bin m_{HH} ; it bins a **multivariate DNN score** (shape systematics propagated by re-running the DNN on varied inputs,

[CMS p. 93]). So CMS already carries the same multi-dimensional systematic information NSBI does — **NSBI is not “exposed to more systematics” than the real CMS baseline.** Against CMS-as-done, the comparison reduces to:

- **What NSBI buys:** *unbinning* (no information lost to binning the score) and a κ_λ -*optimal* per-event ratio (CMS’s DNN is one fixed discriminant, not optimal at every κ_λ). These are the off-shell $\sim 2\times$ / FAIR-Universe $\sim 20\%$ gains.
- **What NSBI costs:** a *learned* morphing layer (NSBI must *train* the per-event systematic weight; CMS *counts* its re-run-DNN template, exact up to MC stat) and the *custom coverage validation*. The danger of “worse than CMS” is the *learned layer being mis-calibrated / under-validated* — producing a tighter-but-mis-covering interval — not a heavier systematics burden.
- **The genuinely new $b\bar{b}\tau\tau$ problem is the data-driven backgrounds** (Sec. 4): if QCD/-fakes cannot be given a faithful per-event density, the unbinned NSBI may have to bin them anyway, partially eroding its advantage in exactly the most sensitive channel.

Protections (so the worst case is “ $\approx$ CMS,” not “worse and unknown”): honest morphing uncertainty (GP-style); explicit coverage validation; and **binning the NSBI discriminant as a built-in cross-check** (the binned fit is a coarsening of NSBI, so a regression is detectable and you fall back).

8 Summary of new challenges

	ATLAS off-shell 4ℓ	κ_λ in $HH \rightarrow b\bar{b}\tau\tau$ (new challenge)
POI	$\mu_{\text{off-shell}}$ (1 POI)	κ_λ ($+k_{2V}, k_V$, EFT); shape in m_{HH}
Final state	4 leptons, fully reco, clean	b -jets $+\tau_h$ +MET; SVfit; 3 channels
Backgrounds	one MC $q\bar{q}ZZ$ (interfering), CR-constrained	QCD (ABCD), DY (18-CR SFs), $t\bar{t}$ (CR-SF), jet $\rightarrow \tau$ fakes , single- H , diboson
Data-driven bkg in NSBI	none needed	QCD & fakes have no per-event density
Experimental systematics	few, small (leptons)	many, large (JES $\times 11$, τ ES, b -tag $\times 5$, DeepTau, ...)
Syst. vs signal shape	largely separable	JES/τES deform m_{HH} (degenerate with κ_λ)
Theory–POI coupling	none	κ_λ - dependent THU _{HH} +4/−18%
Factorisation of NPs	safe	JES $\times \tau$ ES cross-term in m_{HH}
Signal yield	$\sim 15\%$ of σ	$\mathcal{O}(10\text{--}100)$ events (fb-scale)
MC training stats	—	ample ($\approx 500\text{--}750k/\kappa_\lambda$); watch $\pm 1\sigma$ samples
Stats framework	custom (built)	must port the custom fit; CMS has COMBINE

Bottom line

The NSBI *method* ports cleanly from off-shell to κ_λ . The new challenges are in the *inputs*, and rank roughly: (1) **data-driven backgrounds** (QCD, jet $\rightarrow$ τ fakes) with no per-event density — the most awkward fit with NSBI, but *solved-with-caveats* by flow-based estimation (**ABCDnn**, Sec. 4); (2) the κ_λ -**systematic shape degeneracy** (JES/ τ ES on m_{HH}) plus a much larger experimental-systematics surface, which makes *morphing accuracy + honesty* the limiting factor; (3) the κ_λ -**dependent theory** and **factorisation cross-terms** (non-factorisable morphing); (4) **irreducible single-Higgs**. None are NSBI-specific *except* the data-driven-background mismatch and the learned-morphing layer; the rest are equally CMS’s problems, handled with the same NPs. Two scope caveats: NSBI’s gain is in the κ_λ/k_{2V} *parameter constraint* (shape directions), *not* the SM-HH observation significance (rate-saturated, luminosity-set); and the genuine per-event edge is on *shape/correlated-operator* directions, not the rate-broken $\kappa_t \leftrightarrow \kappa_\lambda$ / $k_V \leftrightarrow k_{2V}$ ones.

References

- [1] ATLAS Collaboration (J. Sandesara *et al.*), *Measurement of the Off-Shell Higgs Boson in $H \rightarrow ZZ \rightarrow 4\ell$ using Simulation-Based Inference*, ATLAS note HIGG-2023-01 (2024) = arXiv:2412.01600, Rep. Prog. Phys. 88 (2025). Page refs herein are to the note. Key: mixture model & systematics Sec. 5 (pp. 80–90); per-event factorised weights (pp. 83–84); MC-stat NP (pp. 86–87); JAX fit & impact (pp. 89–90).
- [2] CMS Collaboration (C. Amendola *et al.*), *Search for Higgs boson pairs produced through gluon and vector boson fusion in the $b\bar{b}\tau\tau$ final state with Run-II data*, CMS AN-18-121 (2022). Key: signal reweighting Sec. 3.1 (p. 6); background estimation Sec. 7 (pp. 61–85; QCD ABCD pp. 61–62, DY 18 CRs pp. 63–78, $t\bar{t}$ CR-SF pp. 79–82); systematics Sec. 9 (pp. 90–97).
- [3] S. Choi, J. Lim, H. Oh, *Data-driven Estimation of Background Distribution through Neural Autoregressive Flows (ABCDnn)*, arXiv:2008.03636 (2020). Production application: CMS all-hadronic four-top search, Run-2+3 (ExtendedABCD yield + ABCDnn shape for $t\bar{t}$ +jets/QCD; obs. 3.95σ , $\mu = 2.7^{+1.1}_{-0.9}$) — S. An (advisors M. Paulini, V. Dutta), Carnegie Mellon PhD thesis (2025), CMU KiltHub #29950337; MMD-trained NAF, conditioned on (N_{jets}, N_b) , modelling the BDT+ H_T shape. No public CMS-PAS located.
- [4] A. Bahl, T. Plehn, N. Schmal, *global SMEFT fit of four di-boson processes with per-event likelihoods*, arXiv:2509.05409 (2025) — the $\sim 2\times$ per-event gain on correlated-operator directions.
- [5] ATLAS+CMS HH combination, arXiv:2602.23991 / CMS-PAS-HIG-25-014 (early 2026): $k_{2V} \in [0.73, 1.3]$, $\kappa_\lambda \in [-0.71, 6.1]$, $\mu_{HH} < 2.5$ (95% CL).